

Swollen Lymph Nodes

(Swollen Glands; Lymphadenopathy)

By **James D. Douketis**, MD, McMaster University
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Lymph nodes are tiny, bean-shaped organs that filter [lymph](#). They are located throughout the body, but particular collections are found just under the skin in the neck, under the arms, and in the groin area. Lymph nodes are part of the [lymphatic system](#), which is one of the body's defense mechanisms against the spread of infection and cancer.

Lymph is clear fluid that is made up of water, [white blood cells](#), proteins, and fats that have filtered out of blood vessels into the spaces between cells. Some of the fluid is reabsorbed by the blood vessels, but the rest enters the lymphatic vessels. Lymph then passes through the lymph nodes, which are specific collection points where damaged cells, infectious organisms, and cancer cells are filtered from the fluid and destroyed. If many infectious organisms or cancer cells are present, the lymph nodes swell. Sometimes, organisms cause infection within a lymph node.

Lymphadenopathy is the term doctors use to refer to swollen lymph nodes.

Lymphadenitis is the term used when swollen lymph nodes are painful or have signs of inflammation (for example, redness [which may appear as darkness on people with dark skin] or tenderness), usually due to viral or bacterial infection.

A few small nodes often can be felt in healthy people. Lymph nodes that are larger and easily felt may be a sign of a disorder. Some people use the term "swollen glands" to refer to swollen lymph nodes, especially when the nodes in the neck are swollen. However, lymph nodes are not [glands](#).

Lymph nodes may be swollen in only one body area or in several areas. Other symptoms, such as sore throat, runny nose, or fever, may be present depending on the cause. Sometimes swollen lymph nodes are discovered when the person is being examined because of another symptom.

Causes of Swollen Lymph Nodes

Because lymph nodes participate in the body's [immune response](#), a large number of infections, inflammatory disorders, and cancers are potential causes for a swollen lymph node. Only the more common causes are discussed in detail here. (See the table [Some Causes and Features of Swollen Lymph Nodes](#) for a more complete list.)

Lymph nodes may swell with no known cause and return to normal on their own.

Common causes of swollen lymph nodes include:

- Upper respiratory infections (URI)
- Infections in tissues near the swollen lymph node

Dangerous causes of swollen lymph nodes include:

- Cancer
- [Human immunodeficiency virus](#) (HIV) infection
- [Tuberculosis](#) (TB)

However, most people with swollen lymph nodes do not have these disorders. For example, about 1% of adults with swollen lymph nodes have cancer; however, the risk of cancer is higher (4%) in people older than 40 years.

Evaluation of Swollen Lymph Nodes

Not every person with swollen lymph nodes requires immediate evaluation by a doctor. The following information can help people decide when a doctor's evaluation is needed and help them know what to expect during the evaluation.

Warning signs

In people with swollen lymph nodes, certain symptoms and characteristics are cause for concern. They include:

- A node that is about 2 centimeters (almost an inch) or more in diameter
- A node that is draining pus
- A node that feels hard or does not move around easily under the skin
- Risk factors for HIV infection (such as having been stuck with a needle used by another person or having engaged in high-risk sexual activities) or

tuberculosis (such as living or working with a person who has tuberculosis or having recently moved from an area where tuberculosis is prevalent)

- Fever or unexplained weight loss

When to see a doctor

If a lymph node is very painful or draining pus or other material, people should see a doctor right away. Otherwise, people should call their doctor regarding the swelling. The doctor will decide how quickly they need to be seen based on the presence of warning signs and other symptoms. For people who have no warning signs and otherwise feel well, a delay of a week or so is not harmful.

What the doctor does

The doctor first asks questions about the person's symptoms and medical history. Doctors then do a physical examination. What they find during the history and physical examination often suggests a cause of the swollen lymph nodes and the tests that may need to be done (see table [Some Causes and Features of Swollen Lymph Nodes](#))

Swollen Lymph Nodes



This person with non-Hodgkin lymphoma has swollen lymph nodes on the right side of the neck.

DR P. MARAZZI/SCIENCE PHOTO LIBRARY

Doctors ask:

- Where the swollen nodes are located

- How long the nodes have been swollen
- Whether the person has pain
- Whether the person has recently had an injury (particularly cat scratches or rat bites)
- Whether the person has an infection or symptoms of an infection (for example, a runny nose, cough, fever, sore throat, unexplained weight loss, or tooth or gum pain)
- Whether the person has traveled recently
- What medications the person is taking
- Whether there are symptoms in any other part of the body

Doctors then do a physical examination, checking for fever and examining areas where lymph nodes are found. Doctors check the person for any signs of infection or lumps elsewhere in the body. People who have swollen lymph nodes throughout the body often have a disorder that affects the entire body. However, people who have swollen lymph nodes in only one area may have a disorder that affects only that area (for example, an infection) or more widespread disease.

Sometimes, the history and physical examination findings suggest the cause, as for example when the person has an upper respiratory infection or a dental infection. In other cases, findings do not point to a single cause. People with warning signs are more likely to have a serious disorder, but people with lymph node swelling and no other symptoms may also have a serious disorder.

Nodes that are hard, very enlarged, have been present for longer times (such as several weeks), and do not move when pushed may indicate cancer. Tenderness, discoloration, and warmth in and around a single enlarged lymph node may indicate an infection of the node.



Some Causes and Features of Swollen Lymph Nodes

Cause	Common Features*	Tests
Cancers		
<u>Leukemias</u> (typically chronic and sometimes acute lymphocytic leukemia)	Fatigue, fever, and weight loss With acute leukemia, often easy bruising and/or bleeding	Complete blood count and specialized blood tests (for example, peripheral smear and/or flow cytometry) Bone marrow examination
<u>Lymphomas</u>	Painless lymph node swelling (local or widespread) Nodes often rubbery and sometimes clumped together Often fever, night sweats, and weight loss	Lymph node biopsy Specialized blood tests
Metastatic cancers (often of the head and neck, thyroid, breast, or lung)	One or several painless nodes in the neck Nodes often hard, sometimes unable to be moved when pushed	Tests to identify the primary tumor, often including imaging studies, blood tests, and biopsy
Systemic rheumatic disorders		
<u>Kawasaki disease</u>	Tender swollen nodes in the neck in a child Fever, usually higher than 102° F (39° C), rash on the trunk, prominent red bumps on the tongue, peeling	A doctor's examination, often with blood tests and echocardiogram (heart ultrasound)



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